# Propensity Score Diagnostics

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# Checking balance

- Love plots (Standardized Mean Difference)

- ECDF plots

# Standardized Mean Difference (SMD)

# SMD in R

Calculate standardized mean differences

```r
library(halfmoon)
library(tidyverse)

smds <- tidy_smd(
  df,
  .vars = c(confounder_1, confounder_2, ...),
  .group = exposure,
  .wts = wts # optional,
  make_dummy_vars = TRUE # optional
)
```

# Calculating SMDs

```r
vars <- c("income", "health", "temperature")

smds <- tidy_smd(
  net_data_wts,
  .vars = all_of(vars),
  .group = net,
  .wts = w_ate
)

smds
```

# Calculating SMDs

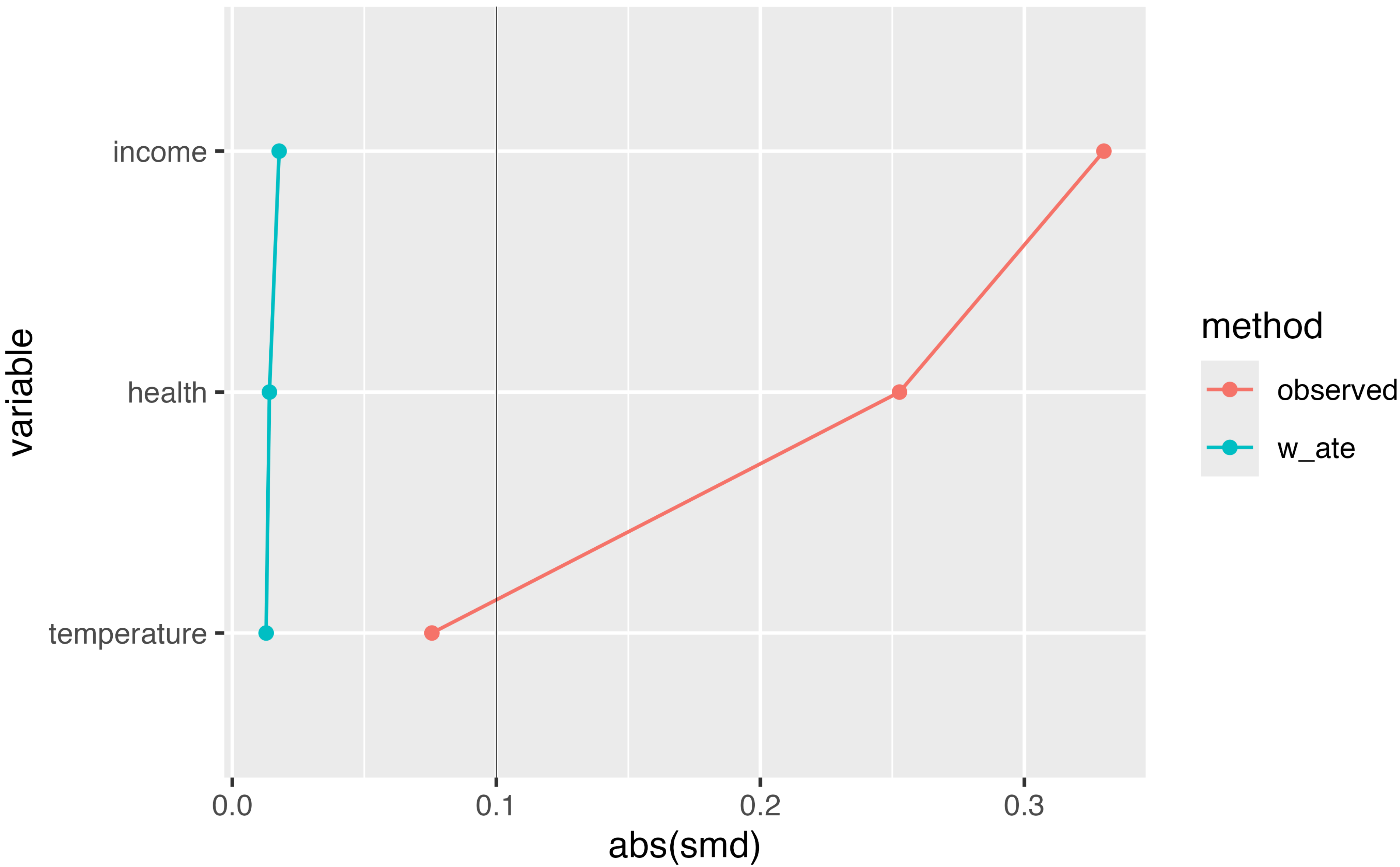
```
# A tibble: 6 × 4
  variable    method   net      smd
  <chr>       <chr>    <chr>  <dbl>
1 income      observed TRUE  -0.330
2 health      observed TRUE  -0.253
3 temperature observed TRUE   0.0756
4 income      w_ate    TRUE   0.0177
5 health      w_ate    TRUE   0.0141
6 temperature w_ate    TRUE  -0.0128
```

# Plotting SMDs

Plot them! (in a Love plot!)

```
ggplot(
  data = smds,
  aes(
    x = abs(smd),
    y = variable,
    group = method,
    color = method
  )
) +
  geom_love()
```

# Love plot



income
health
temperature
variable
0.0
0.1
0.2
0.3
abs(smd)
method
observed
w_ate

# Your turn 1

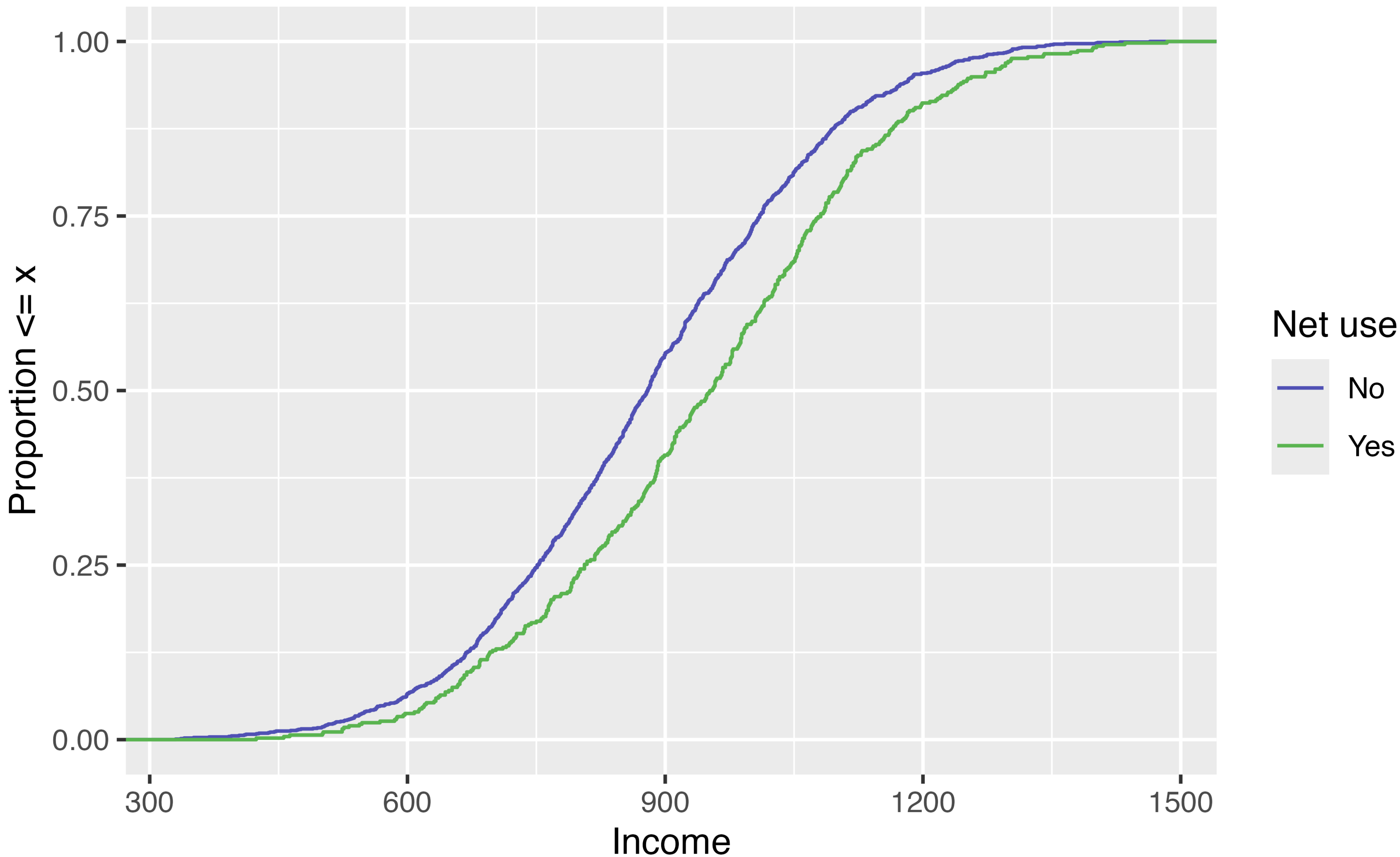
## Create a Love Plot for the propensity score weighting you created in the previous exercise

06:00

# ECDF

For continuous variables, it can be helpful to look at the ***whole*** distribution pre and post-weighting rather than a single summary measure

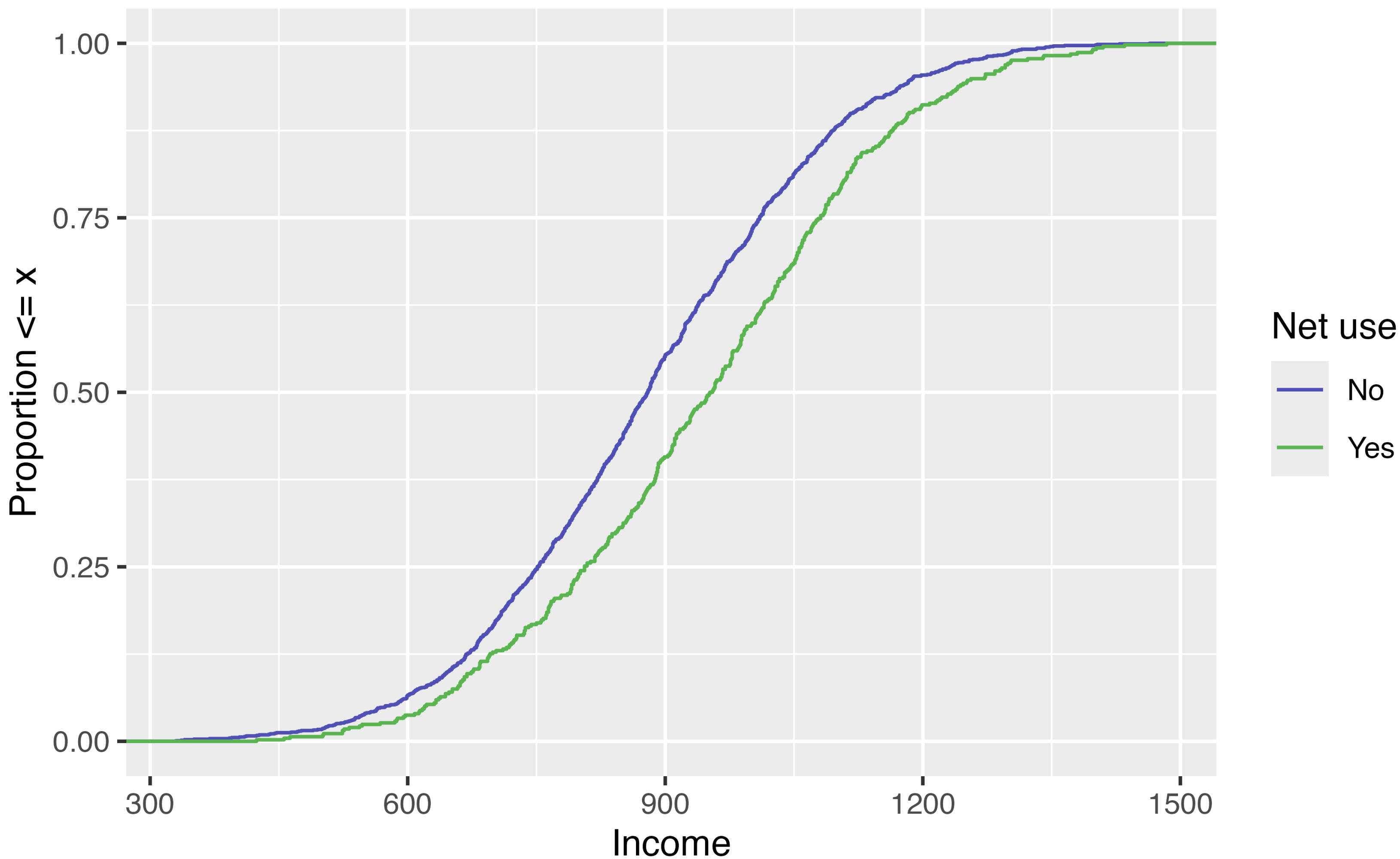
# ECDF


1.00
0.75
0.50
0.25
0.00
Proportion <= x
300
600
900
1200
1500
Income
Net use
No
Yes

# Unweighted ECDF

```r
ggplot(net_data_wts, aes(x = income, color = factor(net))) +
  geom_ecdf() +
  scale_color_manual(
    "Net use",
    values = c("#5154B8", "#5DB854"),
    labels = c("No", "Yes")
  ) +
  xlab("Income") +
  ylab("Proportion <= x")
```

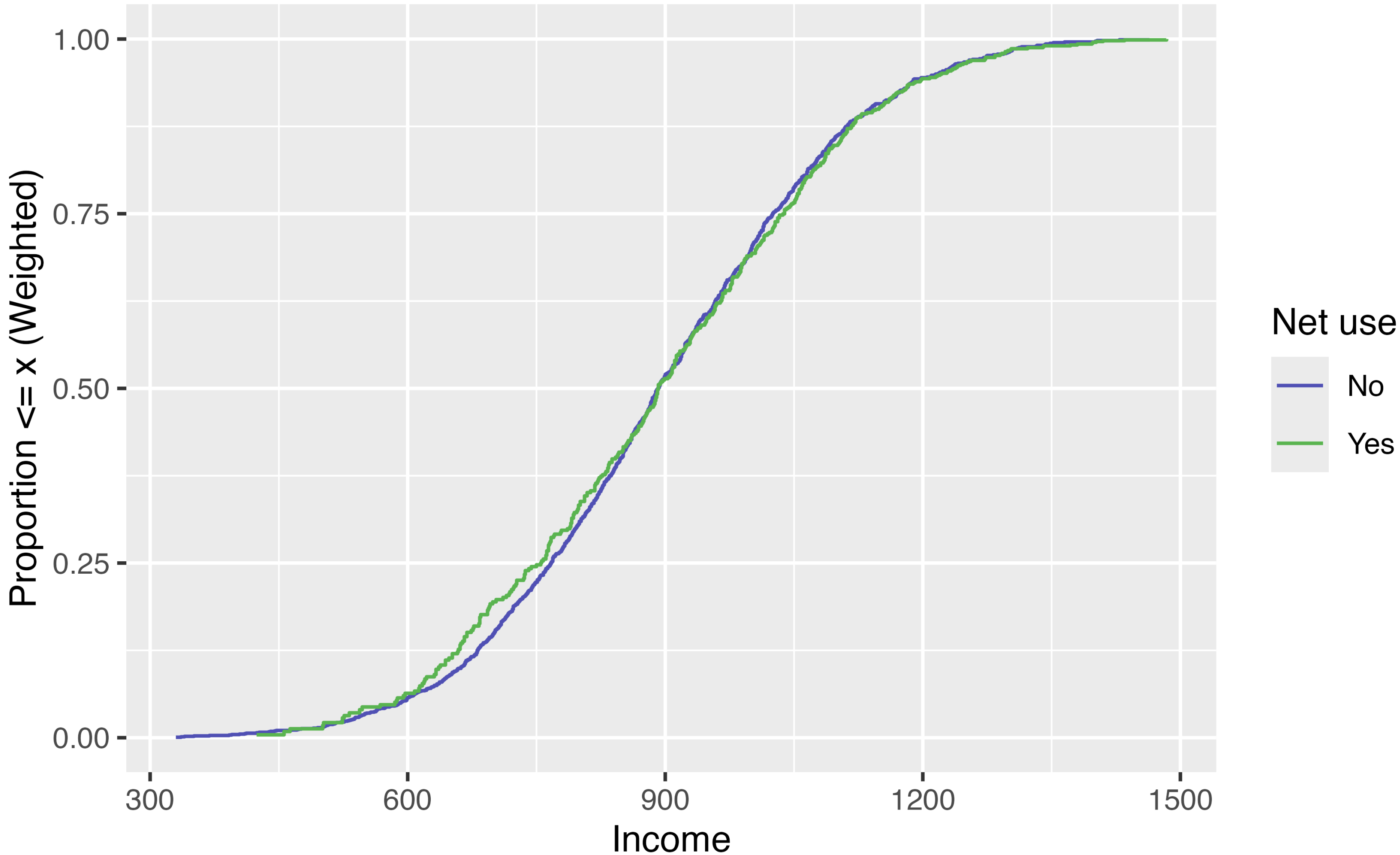
# Unweighted ECDF



# Weighted ECDF

```r
ggplot(net_data_wts, aes(x = income, color = factor(net))) +
  geom_ecdf(aes(weights = w_ate)) +
  scale_color_manual(
    "Net use",
    values = c("#5154B8", "#5DB854"),
    labels = c("No", "Yes")
  ) +
  xlab("Income") +
  ylab("Proportion <= x (Weighted)")
```

# Weighted ECDF



## *Your turn 2*

**Create an unweighted ECDF examining the `park_temperature_high` confounder by whether or not the day had Extra Magic Hours.**

**Create a weighted ECDF examining the `park_temperature_high` confounder**

06:00

# *Bonus!* Weighted Tables in R

# 1. Create a “design object” to incorporate the weights

```r
library(survey)

svy_des <- svydesign(
  ids = ~ 1,
  data = df,
  weights = ~ wts
)
```

# 2. Pass to `gtsummary::tbl_svysummary()`

```r
library(gtsummary)
tbl_svysummary(svy_des, by = x) |>
  add_difference(everything() ~ "smd")
# modify_column_hide(ci) to hide CI column
```

| Characteristic | TRUE<br>N = 1,760[1] | FALSE<br>N = 1,751[1] | Difference[2] | 95% CI[2] |
|---|---|---|---|---|
| income | 892 (755, 1,039) | 892 (767, 1,025) | 0.02 | -0.05, 0.08 |
| health | 49 (37, 64) | 50 (36, 63) | 0.01 | -0.05, 0.08 |
| temperature | 23.8 (20.8, 27.3) | 23.9 (20.9, 27.1) | -0.01 | -0.08, 0.05 |

[1] Median (Q1, Q3)
[2] Standardized Mean Difference
Abbreviation: CI = Confidence Interval

# *Bonus Your Turn*: Weighted Tables